

- 4 (a) Lola is concerned about the risks to her computer when using the Internet.

She wants to use some security methods to help protect her computer from the risks.

Identify a security method she could use for each of the following risks. Each security method must be different.

Describe how each security method will help protect Lola's computer.

- (i) Computer virus

Security method

Description

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.....

.....

[3]

- (ii) Hacking

Security method

Description

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[3]

- (iii) Spyware

Security method

Description

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[3]

6 A law company holds a lot of sensitive data about its clients.

- (a)** It currently requires employees to enter a username and a password to log-in to an account. Each password must be 8 letters.

The company wants to increase the security of the log-in system.

Identify **two** improvements the company could use to make the log-in system more secure.

Explain how each improvement increases security.

Improvement 1

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Explanation

.....

.....

Improvement 2

.....

Explanation

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[4]

8 An art gallery has a website that is used to display and sell art.

- (a)** The gallery uses Secure Socket Layer (SSL) to provide a secure connection when selling art.

Describe the process of SSL and how it provides a secure connection.

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[6]



(b) The art gallery also uses a firewall.

Six statements are given about firewalls.

Tick (✓) to show if the statement is **True** or **False**.

Statement	True (✓)	False (✓)
Firewalls are only available as hardware devices		
Firewalls allow a user to set rules for network traffic		
Firewalls will automatically stop all malicious traffic		
Firewalls only examine traffic entering a network		
Firewalls encrypt all data that is transmitted around a network		
Firewalls can be used to block access to certain websites		

[6]

0478/12 – JUNE 2019

- 5 A music company wants to send a new music file to many radio stations. It will send the music file the day before the release date so that the radio stations can store the file ready for release.

The music company does not want the radio stations to be able to open the music file until 09:00 on the release date.

Identify **two** security measures **and** describe how each measure can be used to make sure the music file cannot be opened until the release date.

Security measure 1

Description

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.....

Security measure 2

Description

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.....

[4]

(b) Priya uses cookies in her website.

Five statements are given about cookies.

Tick (✓) to show if the statement is **True** or **False**.

Statement	True (✓)	False (✓)
Cookies can be used to store a customer's credit card details		
Cookies can be used to track the items a customer has viewed on a website		
Cookies will corrupt the data on a customer's computer		
Cookies are downloaded onto a customer's computer		
Cookies can be deleted from a customer's computer		

[5]

(d) Priya has a URL for her website.

State what is meant by a URL.

.....
..... [1]

(e) Priya is concerned about a denial of service attack (DoS) occurring on her webserver.

(i) Explain what is meant by a denial of service attack.

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.....
..... [4]

(ii) Give **one** security device that can be used to help prevent a denial of service attack.

..... [1]

(d) Customers will use a web browser to access Victoria's website.

Victoria writes a paragraph of text to explain how the website will be displayed on a customer's computer.

Use the list given to complete Victoria's paragraph by inserting the correct **six** missing terms. Not all terms will be used.

- browser
- domain name
- firewall
- hexadecimal
- HTML
- https
- MAC address
- search engine
- Uniform Resource Locator (URL)
- web server

The user enters the website into the address bar.

The protocol that is used is The URL contains

the for the website. This is used to look up the

IP address of the company. A DNS server stores an index of IP addresses.

The browser sends a request to the as this is

where the files for the website are stored. The files are sent back to the

..... as files.

This is interpreted by the browser and the web page is displayed.

[6]

(e) When customers access Victoria's website they will be given the message:

This website uses cookies. An explanation of their purpose can be found in our cookies policy.

(i) Explain what is meant by cookies.

.....

.....

.....

..... [2]

(ii) Explain why Victoria would use cookies as part of her website.

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..... [4]

3 A finance company is concerned that its employees are being distracted by using gaming websites at work.

(a) Explain how a firewall could help prevent this distraction.

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..... [4]

- The company thinks that spyware was used to obtain a user's password.
- Explain how spyware could have been used to obtain the user's password.

[6]

- 3** A company has several offices. It uses the Internet to transfer data between offices. The company also makes payments to staff and suppliers using online banking.

The company are concerned about spyware and other security aspects of using the Internet.

- (a)** Explain what is meant by spyware **and** how it is used to obtain data.

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..... [3]

- (b)** The company uses a web page to log on to the online bank.

Identify **one** method that could be used by the online bank to reduce the impact of spyware when logging on.

State **how** the method prevents the use of spyware.

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..... [2]

- (c)** The company has installed a firewall as part of its data security.

Describe how a firewall can help protect against unauthorised access to data.

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..... [4]

(d) State **two** other methods the company could use to help prevent unauthorised access to data.

Method 1

Method 2

[2]

(b) Ishan shares the program with community members over the Internet, using Secure Socket Layer (SSL).

(i) Explain how Ishan will know he is on a secure website.

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..... [1]

(ii) Describe how an SSL connection is established.

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..... [5]

- (b) Gloria writes a paragraph as an answer to an examination question about accessing a website.

Use the list given to complete Gloria's answer by inserting the correct **four** missing terms. Not all terms will be used.

- browser
- cookies
- Hypertext Markup Language (HTML)
- hypertext transfer protocol (http)
- hypertext transfer protocol secure (https)
- Internet Protocol address (IP address)
- Media Access Control address (MAC address)
- web server

The user enters the URL of the website. The uses the DNS server to look up the of the website.

The browser sends a request to the to obtain the website files. The website files are sent as that is interpreted by the browser.

[4]

(b) State **three** functions of a browser.

- 1
 -
 - 2
 -
 - 3
 -
- [3]

(c) The website Gerald visits uses https.

Explain what is meant by https.

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-
-
- [3]

(b) Companies can use a range of security methods to keep their data secure.

Identify **two** security methods that a company can use to keep their data secure **and** explain how each method can keep the data secure.

Security method 1

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Security method 2

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[6]

0478/13 – NOVEMBER 2019

8 A company discovers malware on its network.

Explain **two** ways that the malware could have been introduced to the company's network.

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[4]



8 A student website provides research support and software downloads.

(a) Students use a browser to access the web pages. Explain the role of a browser in this process.

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..... [5]

(b) The website owners are worried about a denial of service (DoS) attack.

Explain the term denial of service attack.

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..... [3]

(b) Give **three other** methods that can be used to secure the data in the office.

Method 1

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Method 2

.....

Method 3

.....

[3]

5 Meena uses a browser to research information for her business.

(a) Give **three** functions of a browser.

1

2

3

[3]

(b) Meena buys products for her business using the Internet.

The Transport Layer Security (TLS) protocol is used for transferring data when she buys products.

One layer of the TLS protocol is the handshake layer.

(i) Describe the purpose of the handshake layer.

.....

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.....

..... [2]

(ii) Identify the other layer of the TLS protocol.

..... [1]

(iii) Identify another protocol that can be used to transfer data securely.

..... [1]

- (c) Meena visits a website to buy products for her business.

The browser uses a small file to store the details of the products she views. This allows the website to display advertisements for other products she may like.

The small file also stores her log-in details.

Give the name of this type of file.

..... [1]

- 10 Uma is concerned about risks that she may encounter when using the Internet.

Two of the risks she is concerned about are phishing and pharming.

- (a) Give **one** similarity and **two** differences between phishing and pharming.

Similarity

.....

.....

Difference 1

.....

.....

Difference 2

.....

.....

[3]

- (b) Identify **two** other risks that Uma could encounter when using the Internet.

Risk 1

Risk 2

[2]

- (c) Uma uses a firewall to secure the data on her computer.

- (i) Uma tells her friend that a firewall can only be software-based.

Tick (✓) to show whether Uma is **Correct** or **Incorrect**.

☐ **Correct**

☐ **Incorrect**

[1]

(ii) Describe how the firewall helps to keep Uma's data secure.

.....

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.....

.....

.....

0478/13 – JUNE 2020

3 Thomas has an online business that sells homemade furniture. He has a web server that hosts his website for his business.

(a) Describe the role of a web browser in requesting and displaying the web pages for the website.

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..... [3]

(b) Thomas is worried about a denial of service (DoS) attack on his web server.

Describe what happens in a denial of service attack.

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..... [3]

- 4 The table shows **four** definitions.

Complete the table giving the missing **Term** for each definition.

Term	Definition
	A data transmission method that sends data one bit at a time, down a single wire
	An address given to a device on a network. The address is assigned by the network
	The software used to render HTML and display a web page
	An address given to a device at the manufacturing stage that can be used to identify the device on a network

[4]

- 7 Edie uses a firewall to help prevent her children from accessing websites that she does not want them to see.

- (a) Describe how the firewall helps prevent her children from accessing these websites.

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..... [4]

- (b) Edie is concerned that her children may download a virus when accessing websites.

State what is meant by a **virus** and explain what could happen if a virus was downloaded.

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..... [3]

(c) Edie explains to her children how to identify if a website is secure.

(i) Give **two** ways that her children can identify if a website is secure.

1

2

[2]

(ii) Describe how a browser checks that a website is secure.

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..... [4]

10 Clive has a laptop computer that he uses for his business. He enters a username and password to log in to his laptop.

Clive is worried about spyware being used to find out his username and password.

(a) Describe how spyware could be used to find out Clive's username and password.

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.....

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.....

.....

..... [4]

(b) The threat of spyware makes Clive concerned about typing a password to log in to his laptop.

Give an example of how Clive could log in securely without typing a password.

..... [1]



- 3 Transport Layer Security (TLS) protocol is used to secure the transmission of data over the Internet.

(a) Identify the **two** layers in the TLS protocol.

Layer 1

Layer 2

[2]

(b) The following paragraph explains how data is sent securely using the TLS protocol.

Use the terms to complete the paragraph. Not all terms may need to be used.

- authentic
- binary
- browser
- certificate
- internet service provider
- signal
- web page
- web server
- website

The browser requests the to identify itself by providing its This is sent and a check is performed to see if it is If it is, the sends a back to the web server and data transmission begins.

[5]

(c) Identify **one** other protocol that can be used to secure data transmission over the Internet.

..... [1]

(b) Luke is concerned that his web server may be hacked or subjected to a denial of service (DoS) attack.

State **two** security methods that Luke could use to help protect the web server from hacking or a DoS attack.

Method 1

Method 2

[2]

- (i) Identify the **two** layers that are present in the TLS protocol.

Layer 2

(ii) Explain how data is sent securely using the TLS protocol.

..... [6]

(e) Tina is concerned about security threats to her web server.

(i) Identify **three** security threats to her web server that Tina might be concerned about.

1

2

3 [3]

(ii) Tina installs a proxy server to help protect her website from security threats.

Describe how the proxy server will help protect the website.

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..... [4]

(c) Tammy uses a browser when accessing the Internet.

Describe the role of the browser.

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..... [4]



- 4 Eugene has a web server that stores his online shopping website.

Customers access the website using a browser.

- (a) Describe how the webpages are requested and displayed on the customer's computer.

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..... [4]

- (b) State **three** online security threats to Eugene's web server.

Threat 1

Threat 2

Threat 3 [3]

- (c) Arjun uses the Internet to send the digital documents to his friend. He wants to make sure the documents are sent securely.

Identify **two** protocols that can be used to transfer data securely.

Protocol 1

Protocol 2 [2]

13 Phishing and pharming are two security issues a user should be aware of when using the Internet.

(a) State **one** similarity between phishing and pharming.

.....
..... [1]

(b) Explain **two** differences between phishing and pharming.

Difference 1

.....
.....

Difference 2

.....
..... [2]

0478/11 – MARCH 2021

(c) Users can buy the high definition photographs from the website. When a user buys a high definition photograph, a Secure Socket Layer (SSL) connection is created.

(i) Give **one** benefit of using an SSL connection.

.....
..... [1]

(ii) Explain how the SSL connection is created.

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.....
.....
..... [4]

6 Hacking is one type of Internet risk used to obtain personal data that is stored on a computer.

(a) Explain how a firewall can help prevent hacking.

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.....

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..... [4]

(b) Identify and describe **two** other types of internet risk that are used to obtain personal data.

Internet risk 1

Description

.....

.....

.....

Internet risk 2

Description

.....

.....

..... [6]

3 A firewall can be used to help keep the data secure that is stored on a computer.

(a) The given paragraph describes how the firewall operates to help keep the data secure.

Complete the paragraph using the most appropriate terms from the given list. **Not** all of the terms on the list need to be used.

- Accept
- Criteria
- Hacking
- Input
- Network
- Outgoing
- Output
- Processor
- Reject
- Software
- Store
- Storage

A firewall can be or hardware based. It monitors traffic between the computer and the The user sets for the traffic. The firewall will or the traffic based on this. It can help prevent and malicious software that could be a threat to the security of the data.

[6]

(b) Identify **three** other methods that could be used to keep the data secure.

Method 1

Method 2

Method 3

[3]

4 Two internet risks are phishing and pharming.

Describe what is meant by phishing and pharming.

Phishing

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.....

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.....

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Pharming

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[6]

(b) Describe how the web pages for the website are requested and displayed on a user's computer.

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[4]

(c) The web server that hosts the website suffers a denial of service (DoS) attack.

Explain why this will prevent users from accessing the website.

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[2]



- 4 Three types of Internet security risk are virus, spyware and denial of service (DoS) attack.

(a) Six statements are given about Internet security risks.

Tick (✓) to show whether the statement applies to virus, spyware or denial of service. Some statements may apply to more than one Internet security risk.

Statement	Virus (✓)	Spyware (✓)	Denial of service (✓)
captures all data entered using a keyboard			
can be installed onto a web server			
prevents access to a website			
is malicious code on a computer			
is self-replicating			
damages the files on a user's hard drive			

[6]

(b) Identify **three** other types of Internet security risks.

- 1
- 2
- 3

[3]

(c) Some Internet security risks can maliciously damage data. Data can also be damaged accidentally.

State **three** ways that data could be accidentally damaged.

- 1
- 2
- 3

[3]

6 Cookies can be used to store a user's personal data and online browsing habits.

- (a)** A cookie could be used to automatically enter a user's payment details when the user makes a purchase online.

Describe how cookies can be used to store and automatically enter a user's payment details.

.....

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..... [3]

- (b)** Explain why a user may be concerned about their personal data and online browsing habits being stored in cookies.

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..... [4]

- 4 Spencer finds out that his online music account has been accessed by an unauthorised person.
He believes his personal details for the account were obtained using phishing.

(a) Explain how the personal details could have been obtained using phishing.

.....

.....

.....

.....

.....

..... [3]

(b) Give **two** other Internet security risks that could have been used to obtain the personal details.

1

2 [2]

NOVEMBER 2021 (1)

- 11 The table contains descriptions relating to web pages and the Internet.

Complete the table with the correct terms for the given descriptions.

Term	Description
	the language used to create a web page
	the type of software application used to display a web page
	an address given to a computer, by a network, to allow the computer to be uniquely identified
	a text file sent by a web server to collect data about a user's browsing habits
	the company that provides a connection to the Internet

[5]

NOVEMBER 2021 (2)

- 5 Sammi works for a finance company and has a laptop that he uses for his work. He has confidential data about his customers stored on his laptop.

Sammi does **not** connect the laptop to any networks.

- (a) Sammi is concerned about his customers' confidential data being viewed by other people in his office.

One method he uses to prevent others viewing the data is encryption.

Identify **three** other methods Sammi could use to prevent his customers' confidential data being viewed.

1

2

3

[3]

10 Jermain uses the Secure Socket Layer (SSL) protocol for secure transmission when sending data using the internet.

(a) Explain how the SSL protocol secures the data for transmission.

.....

.....

.....

..... [2]

(b) Identify an alternative protocol that could be used for secure transmission of data using the internet.

..... [1]

(c) Give **two** ways that a user can identify if a website uses secure data transmission.

1

.....

2

.....

[2]

7(d) Julius makes sure that all data transmission to and from his website is secure.

One way a user can check if his website uses secure data transmission is to check its certificate.

Give **two** other ways that a user can identify if his website uses secure data transmission.

1

.....

2

.....

[2]

10 (a) A denial of service (DoS) attack is a type of Internet security risk.

State the purpose of a denial of service attack.

.....

..... [1]

(b) Phishing and pharming are also types of Internet security risk. They have the same purpose.

State the purpose of phishing and pharming.

.....

..... [1]

(c) Identify **three** other types of Internet security risk.

1

2

3

[3]

5 (b) The website is stored on a web server. Customers use their own computers to access the website.

Uniform Resource Locator (URL), Internet Protocol (IP) and Media Access Control (MAC) are three types of address used by computers when accessing the Internet.

Give **two** features of each type of address in the table.

Type of address	Features
URL	
IP	
MAC	

[6]

5 (c) The company is concerned about threats to its web server.

Complete the table by identifying:

- **two** potential threats to its web server
- the impact each threat could have on the company
- software the company can use to help to limit each threat.

Each impact must be different.

Threat	Impact on company	Software
.....		
.....		

[6]

- 7 Adele chooses to set a biometric password for her mobile device, instead of a personal identification number (PIN).

(a) State what is meant by a biometric password.

.....
..... [1]

(b) Give **two** reasons why a biometric password is more secure than a PIN.

Reason 1
.....
Reason 2
..... [2]

(c) Adele has a software-based firewall installed on her mobile device.

The firewall gathers data about the traffic coming into and going out of her mobile device.

Explain how the firewall uses the gathered data to keep the mobile device more secure.

.....
.....
.....
.....
.....
..... [3]

(d) Adele also encrypts the data on her mobile device to keep it more secure.

State how encryption will keep the data more secure.

.....
..... [1]

- 9 Three Internet terms are browser, Internet Protocol (IP) address and Uniform Resource Locator (URL).

Five statements are given about the Internet terms.

Tick (✓) to show which statements apply to each Internet term. Some statements may apply to more than **one** Internet term.

Statement	Browser (✓)	IP address (✓)	URL (✓)
it contains the domain name			
it is a type of software			
it converts Hypertext Markup Language (HTML) to display web pages			
it is a type of address			
it stores cookies			

[5]

3 Joelle is a student who uses the Internet.

(a) The table contains **five** terms or definitions that relate to the Internet.

Complete the table by writing each missing term or definition.

Term	Definition
browser	
.....	this is the company that provides a user with a connection to the Internet
.....	this is a protocol that is used to send data for web pages across the Internet
Uniform Resource Locator (URL)	
cookie	

[5]

- (b)** Joelle uses a firewall to keep her data safe when she uses the Internet.

Tick (✓) to show which statement about firewalls is true.

Tick (✓)

Firewalls can only be hardware-based

9

Firewalls can only be software-based

9

Firewalls can be hardware-based or software-based

9

[1]

- (c)** Joelle's parent also uses the firewall to limit the websites that Joelle can access.

Explain how the firewall is used to limit the websites that Joelle can access.

[4]

[4]

- 6** Millions of emails are sent between users on a daily basis.

- (a) Identify **two** online security attacks that can be carried out using email.

Describe how email is used to enable the attack.

Online security attack 1

Description

.....

.....

.....

.....

Online security attack 2

Description

.....

.....

.....

.....

[6]

(b) Online security attacks can maliciously damage data.

One security method to keep data safe from online attacks is a firewall.

Identify **two** other security methods that keep data safe from online attacks.

Security method 1

Security method 2

[2]



4 Freddie types a Uniform Resource Locator (URL) into the address bar of his browser.

(a) A URL is a text-based version of a website address. It is made up of several different parts.

State the names of **three** different parts of a URL.

Part 1

Part 2

Part 3 [3]

(b) Explain how the URL is converted into an Internet Protocol (IP) address for the website.

.....
.....
.....
..... [2]

(c) A protocol is used to securely transmit the data for the website to the browser.

(i) State the name of this protocol.

..... [1]

(ii) Identify how this protocol changes the data to transmit the data securely.

..... [1]

1 Computers can be infected with malware. Spyware is one example of malware.

(a) Tick (✓) **one** box to show a correct definition of spyware.

A Software that activates a webcam and transmits the video to a third party that outputs it live on a website.

☐

B Software that detects when a password is being entered and then emails the password to a third party.

☐

C Software that records all data entered into a computer, analyses this data to find email addresses and passwords, then posts these to a website.

☐

D Software that records all key presses and transmits these to a third party.

☐

[1]

(b) Complete the table by identifying and describing **two** other examples of malware.

	Malware	Description
1		
2		

[6]

(c) Proxy-servers and firewalls have some similar functions.

Identify **two** similarities and **one** difference between proxy-servers and firewalls.

Similarity 1

.....

.....

Similarity 2

.....

.....

Difference

.....

.....

[3]

(d) The website allows the user to set up an account to log on and purchase items. The website is accessed and displayed using a web browser.

(i) Two functions of the web browser are to render hypertext markup language (HTML) to display web pages and to store cookies.

Identify **two** other functions of a web browser.

1

2

[2]

(ii) Identify **two** ways that cookies can be used to enhance the user's experience of this website.

1

.....

2

.....

[2]



- 8 A manager at a company is concerned about a brute-force attack on its employee user accounts.
- (a) Describe how a brute-force attack can be used to gain access to the employee user accounts.
-
-
-
-
-
- [3]
- (b) One possible aim for carrying out a brute-force attack is to install malware onto the company network.
- (i) State **two** other aims for carrying out a brute-force attack to gain access to the employee user accounts.
- 1
-
- 2
- [2]
- (ii) Identify **three** types of malware that could be installed.
- 1
- 2
- 3
- [3]
- (c) Give **two** security solutions that could be used to help prevent a brute-force attack being successful.
- 1
-
- 2
- [2]

10 A student uses the internet for their schoolwork to research what is meant by pharming.

(a) State the aim of pharming.

.....
..... [1]

(b) Draw and annotate a diagram to represent the process of pharming.

[4]

(c) The student uses a web browser to access data on the internet.

Explain the purpose of the web browser.

.....
.....
.....
..... [2]

(d) Storing cookies is one function of the web browser.

Give **three** other functions of the web browser.

- 1
-
- 2
-
- 3
-

[3]

(e) A student visits a website that uses session cookies, instead of persistent cookies.

Explain the difference between session cookies and persistent cookies.

-
-
-
-
-
-
-
-
-
-

[4]

6 (a) Complete the statements about cookies.

Use the terms from the list.

Some of the terms in the list will **not** be used. Some terms may be used more than once.

compression executable hypertext markup language (HTML)

hypertext transfer protocol (HTTP) image internet protocol (IP) address

persistent session sound text

uniform resource locator (URL) web browser web server

Cookies are small files that are sent between a
..... and a
..... cookies are stored in memory and **not** in the user's
secondary storage.

When the web browser is closed a cookie is lost,
whereas a cookie is **not** lost.

[6]

(b) Give **three** functions of a cookie.

- 1
- 2
- 3

[3]

7 A distributed denial of service attack (DDoS) is a cyber security threat.

(a) Draw and annotate a diagram to represent the process of a DDoS.

[6]

(b) State **two** aims of carrying out a DDoS attack.

1

.....

2

.....

[2]

(c) Give **two** security solutions that can be used to help prevent a DDoS attack being successful.

1

.....

2

.....

[2]

6 A student is writing a help guide about how to recognise and avoid the cyber-security threat of pharming.

(a) Give **three** appropriate solutions he could include.

- 1
-
- 2
-
- 3
-

[3]

(b) The student also wants to include information in the help guide about the use of social engineering as a cyber-security threat.

Describe what is meant by social engineering.

Include **one** example of social engineering in your answer.

.....

.....

.....

.....

.....

.....

..... [3]

(c) The student includes information about the security solution of access levels.

Describe what is meant by access levels.

.....

.....

.....

.....

.....

..... [3]

8 Complete the statements about a distributed denial of service (DDoS) attack.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

anti-virus	bot	botnet	hacker
internet	malware	secondary storage	
spyware	web browser	web server	website

The attacker encourages people to download onto their computer. This will turn each computer into a , creating a network called a

When the attacker wants the DDoS to take place, repeated requests are simultaneously sent from the computers to a This causes it to crash, meaning that users can no longer access the that is stored on this hardware.

[5]

6 Draw and annotate a diagram to demonstrate how a firewall works.

[4]

1 Malware can be used to corrupt data stored on a computer.

(a) Tick (✓) **one** box to show which cyber security threat is **not** a type of malware.

A Phishing

☐

B Ransomware

☐

C Virus

☐

D Worm

☐

[1]

(b) Identify **one** other example of malware than those given in **part 1(a)**.

..... [1]

(c) Identify the type of software that is used to find and remove malware from a computer.

..... [1]

8 (a) Draw and annotate a diagram that demonstrates the cyber security threat of data interception.

[4]

(b) Identify **one** security solution that will help keep data safe from data interception and state why it will help keep the data safe.

.....

.....

.....

..... [2]



9 The table contains terms and descriptions about the internet.

Complete the table with the missing terms and descriptions.

Term	Description
.....	the collective name for all the web pages available
.....	a small text file, stored by the web browser, that can store a user's personal data
uniform resource locator (URL)	
web server	
.....	the language used to create a website. Example tags are <head> and <body>
.....	a protocol that is used to request and send web pages

[6]

12 Digital currency can be used to pay for products and services.

Digital currencies are often tracked using digital ledgers.

(a) Give **two** other features of digital currency.

1

.....

2

.....

[2]

(b) Identify the process that uses a digital ledger to track the use of digital currency.

..... [1]

4 An employee uses a web browser on their computer.

(a) Describe the main purpose of a web browser.

.....

.....

.....

..... [2]

(b) The employee wants his payment details to be automatically filled in when he buys products using the internet.

Identify the function of a web browser that could be used for this purpose.

..... [1]

(c) The employee wants to be able to quickly access websites that he regularly uses.

Identify the function of a web browser that could be used for this purpose.

..... [1]

(d) The web browser uses the secure socket layer (SSL) protocol to transmit personal data securely over the internet.

State how the SSL protocol secures the data for transmission.

.....

..... [1]

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